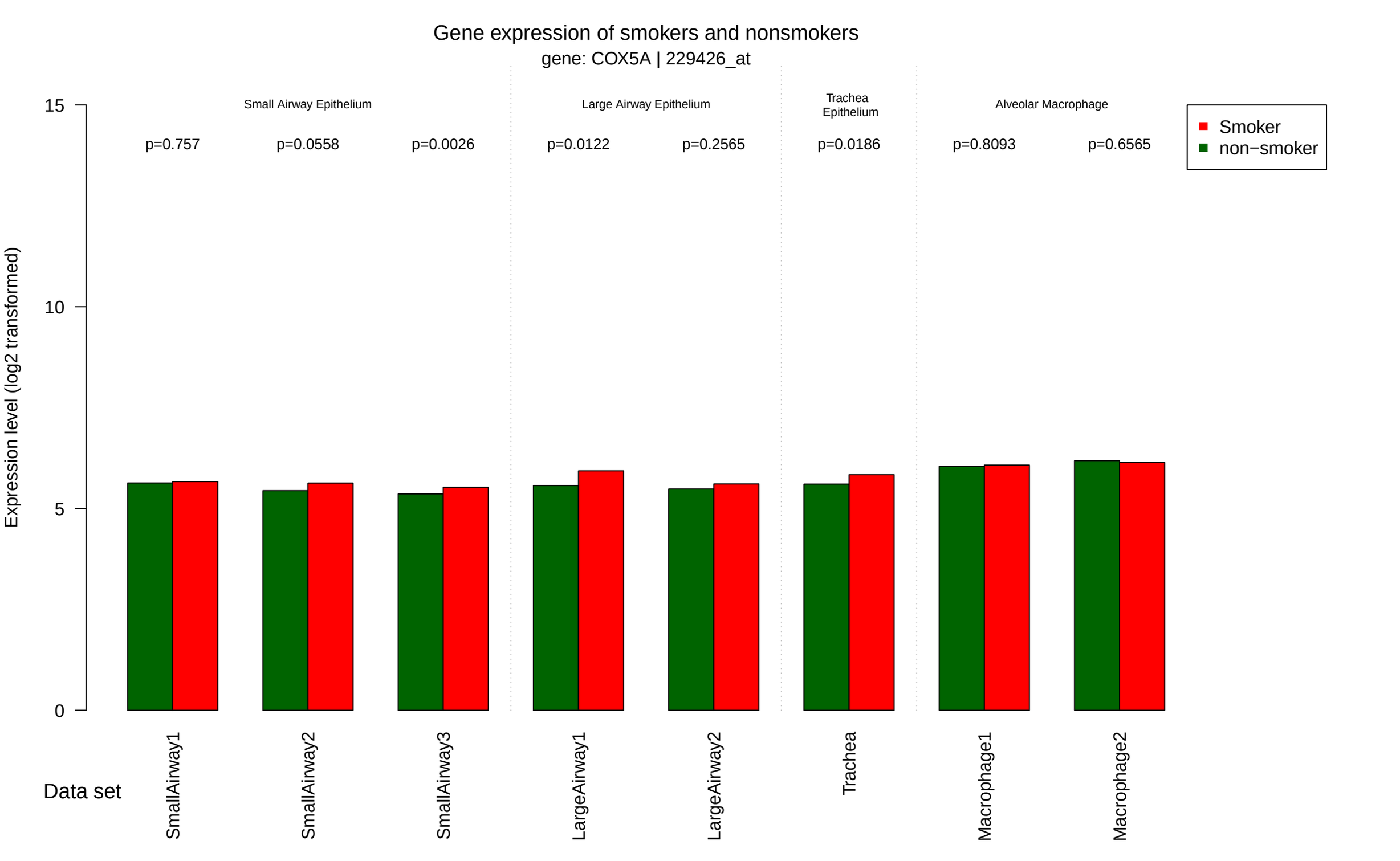
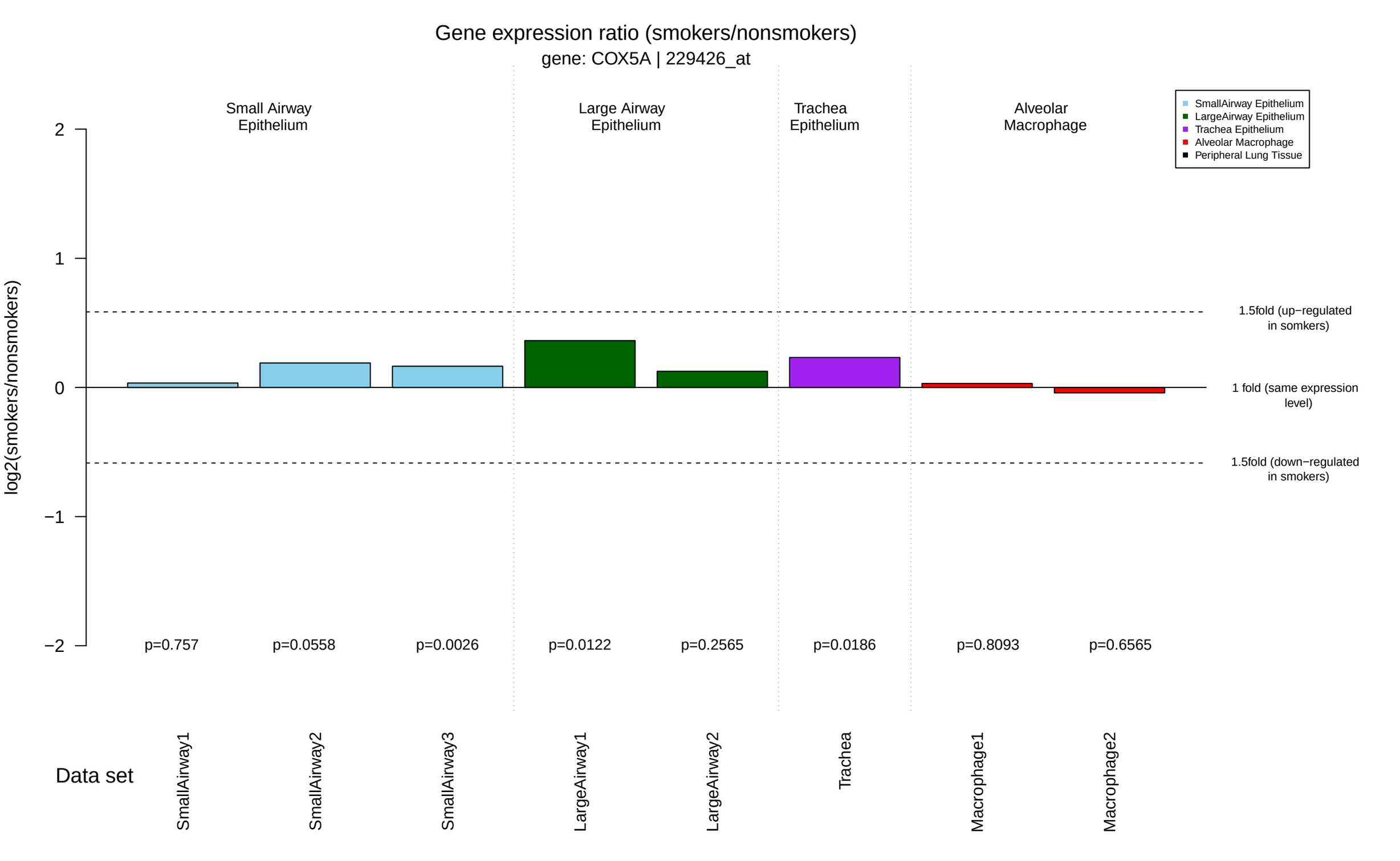
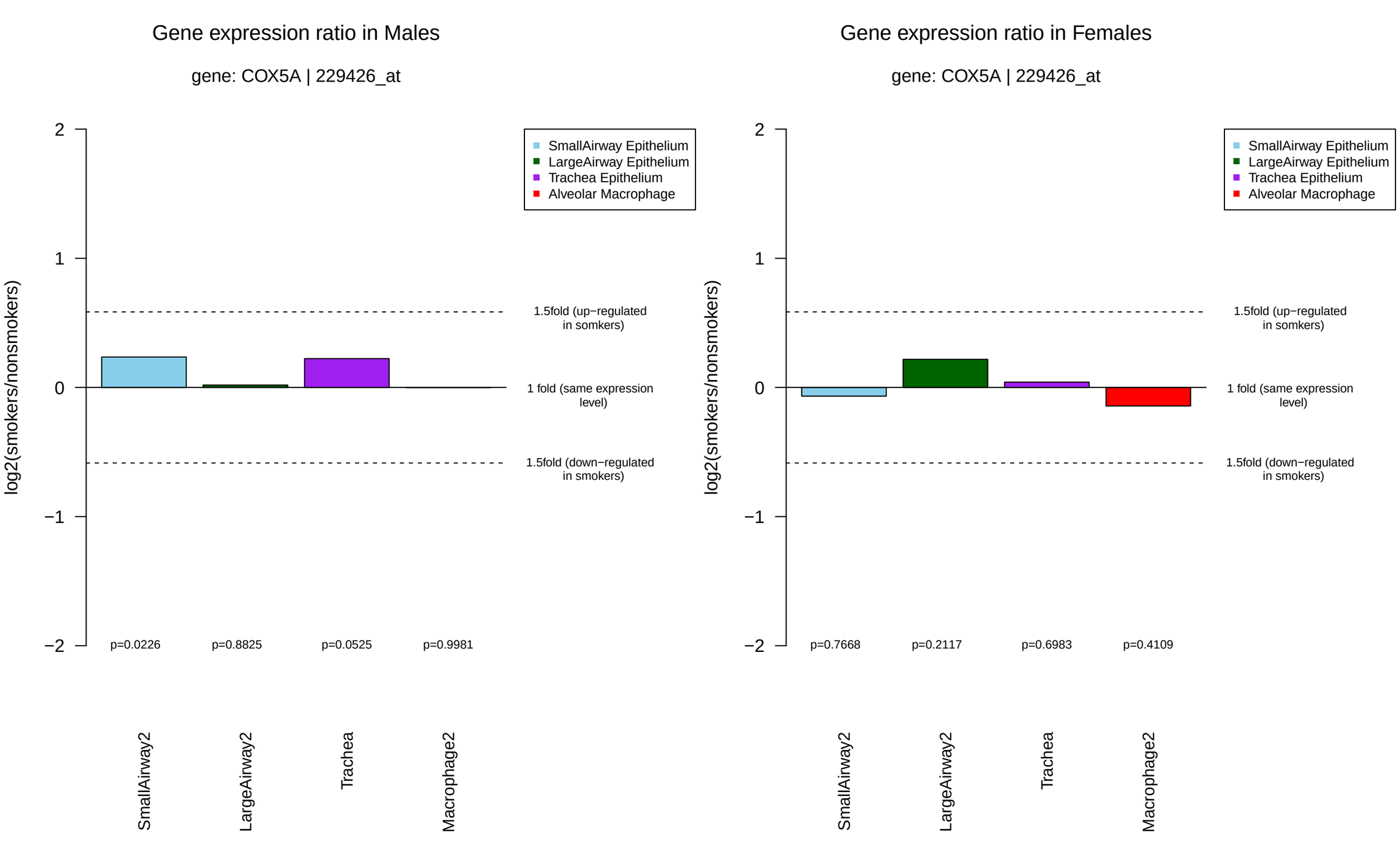
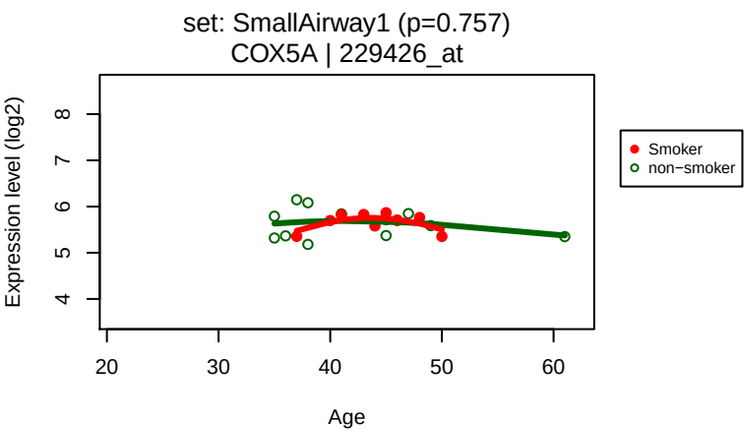
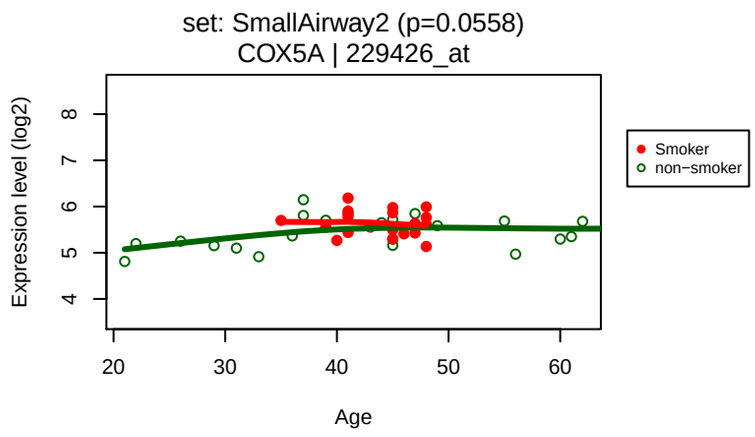
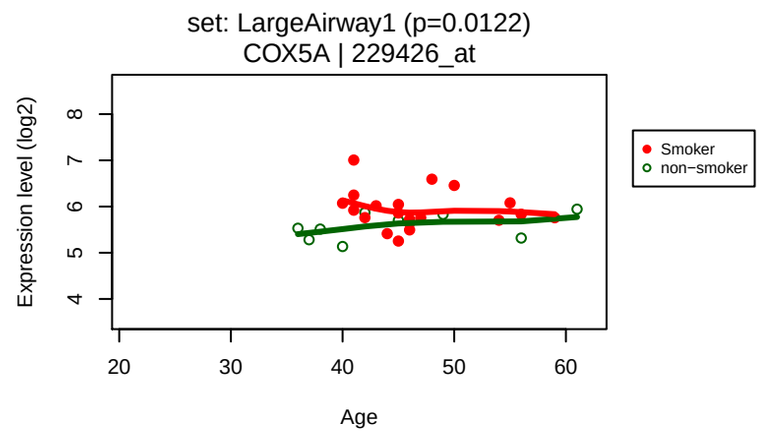

Gene expression of smokers and nonsmokers
gene: COX5A | 229426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.757
p=0.0558
p=0.0026
p=0.0122
p=0.2565
p=0.0186
p=0.8093
p=0.6565
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

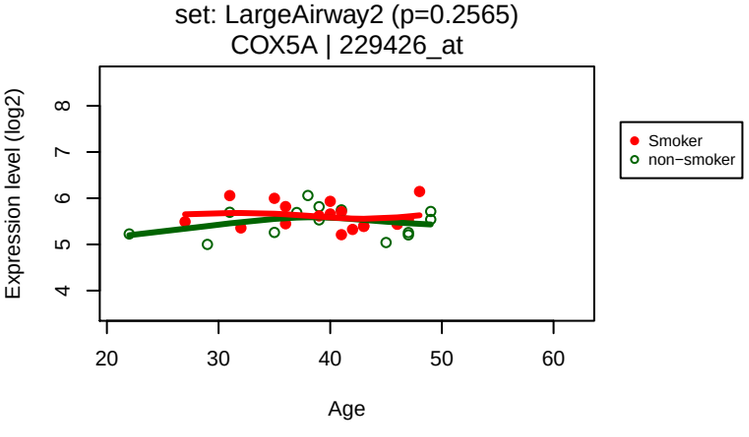

Gene expression ratio (smokers/nonsmokers)
gene: COX5A | 229426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.757
p=0.0558
p=0.0026
p=0.0122
p=0.2565
p=0.0186
p=0.8093
p=0.6565
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

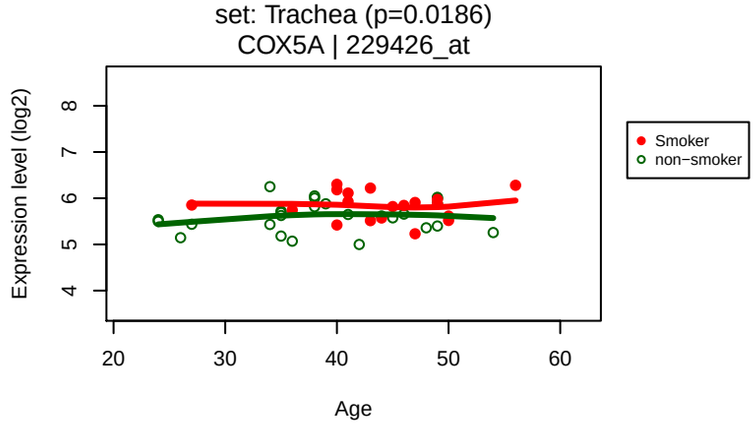

Gene expression ratio in Males
gene: COX5A | 229426_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0226
p=0.8825
p=0.0525
p=0.9981
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX5A | 229426_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7668
p=0.2117
p=0.6983
p=0.4109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

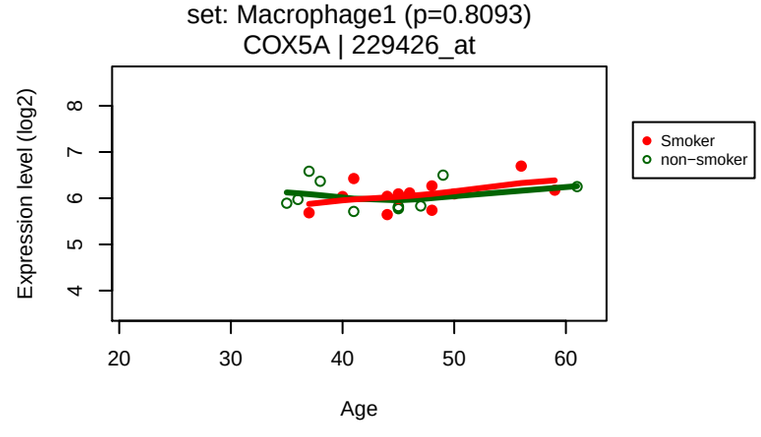

set: SmallAirway1 (p=0.757)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker

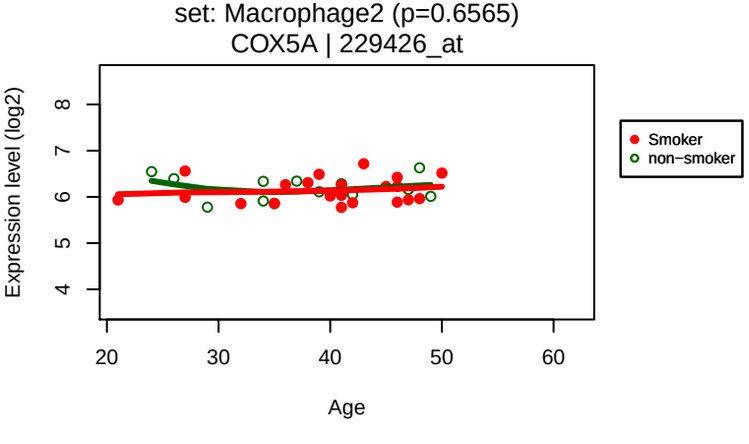

set: SmallAirway2 (p=0.0558)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker

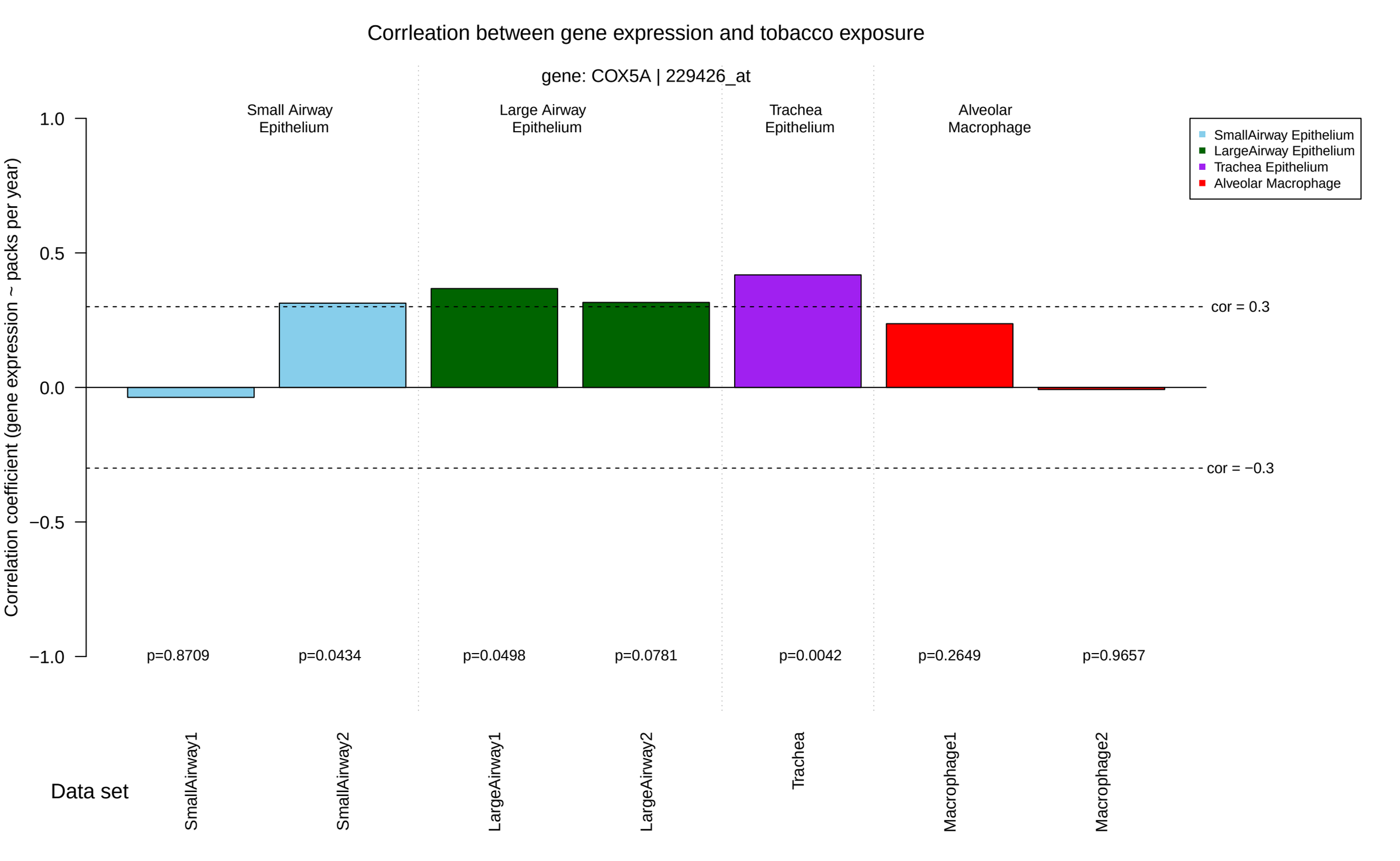

set: LargeAirway1 (p=0.0122)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2565)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0186)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8093)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6565)
COX5A | 229426_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX5A | 229426_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8709
p=0.0434
p=0.0498
p=0.0781
p=0.0042
p=0.2649
p=0.9657
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2